

DS	Session	Arrays	Question Information	Level 1	Challenge 25
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Problem Description:

For some reason, your school's football team has chosen to spell out the numbers on their jerseys instead of using the usual digits. Being great fans, you're going to be ready to cheer for your favorite players by bringing letter cards so you can spell out their number. Each fan has different favorites, so they each need to bring different sets of letters.

The English spellings for the numbers 0 to 12 are:

ZERO ONE TWO THREE FOUR FIVE SIX
SEVEN EIGHT NINE TEN ELEVEN TWELVE

Input Format:

Read a set of integers from 0 to 12, separated by spaces, representing one fan's favorite players. The last integer will be 999, marking the end of the line.

Output Format:

Print the same numbers, then a period and a space. Then, in alphabetical order, print all the letters the fan needs to be able to spell any one of the jersey numbers provided

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 2 3 0999	INPUT (STDIN) 8 14 0999

EXPECTED OUTPUT

2 3 0999. E E H O R T T W

EXPECTED OUTPUT

8 14 0999. E G H I T

▼ Mandatory Test Cases

Test Case 1

KEYWORD

char nums[13][256]

Test Case 2

KEYWORD

for(n=0;n<26;n++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

270

Test Case 3

NLOC

40

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char nums[13][256] = {
6         "ZERO", "ONE", "TWO", "THREE", "FOUR", "FIVE", "SIX",
7         "SEVEN", "EIGHT", "NINE", "TEN", "ELEVEN", "TWELVE"
8     };
9     int val;
10    int inputs[100];
11    int count = 0;
12
13    while (scanf("%d", &val) != EOF) {
14        inputs[count++] = val;
15        if (val == 999) {
16            int counts[26] = {0};
17            for (int i = 0; i < count - 1; i++) {
18                int num = inputs[i];
19                if (num >= 0 && num <= 12) {
20                    for (int j = 0; nums[num][j] != '\0'; j++) {
21                        counts[nums[num][j] - 'A']++;
22                    }
23                }
24            }
25        }
26    }
```

```
26         if (inputs[i] == 999) {
27             printf("0999. ");
28         } else {
29             printf("%d ", inputs[i]);
30         }
31     }
32     int first = 1;
33     int n;
34     for (n = 0; n < 26; n++) {
35         while (counts[n] > 0) {
36             if (!first) printf(" ");
37             printf("%c", 'A' + n);
```

```
38         first = 0;
39         counts[n]--;
40     }
41 }
42 printf("\n");
43 count = 0;
44 }
45 }
46 return 0;
47}
48
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
2 3 0999
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array5.exe
```

```
2 3 0999. E E H O R T T W
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
8 14 0999
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array5.exe
```

```
8 14 0999. E G H I T
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```